

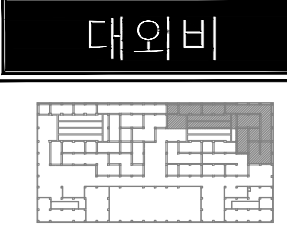
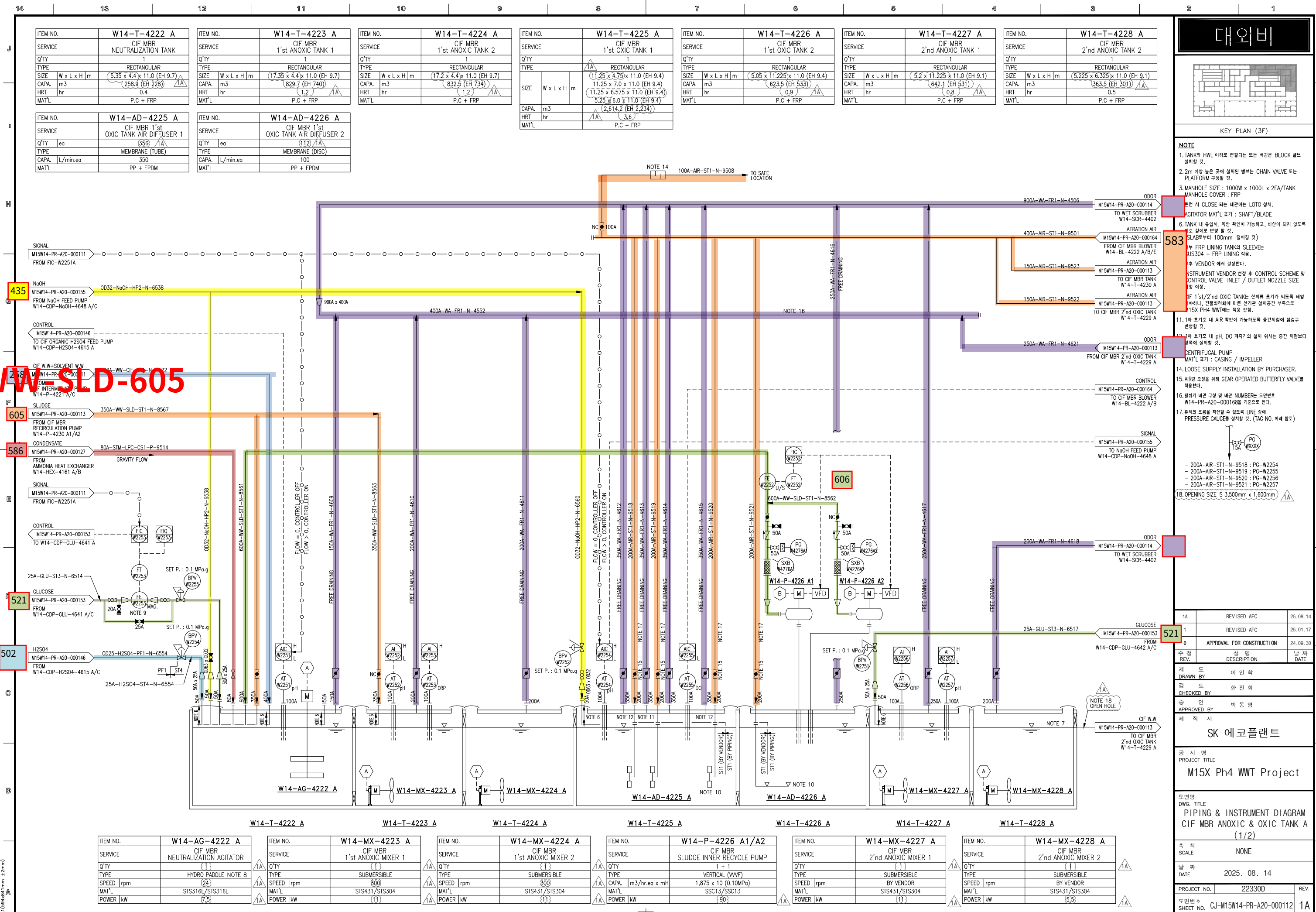
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

WW-SLD-605

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	605	3FA	350A-WW-SLD-ST1-N-8567-01	1 OF 1	1	CJ-M15W14-PR-A20-000113	1A	
2	605	3FA	350A-WW-SLD-ST1-N-8568-01	1 OF 1	1	CJ-M15W14-PR-A20-000113	1A	
3	605	4FA	350A-WW-SLD-ST1-N-8563-01	1 OF 1	0	CJ-M15W14-PR-A20-000112	1A	
4	605	4FA	350A-WW-SLD-ST1-N-8567-01	1 OF 1	0	CJ-M15W14-PR-A20-000113	1A	



KEY PLAN (3F)

- NOTE**
1. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 2. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 3. MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP
 4. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
 5. AGITATOR MAT'L 표기 : SHAFT/BLADE
 6. TANK 내 유입시, 유출시, 유입이 가능하고, 배관이 되지 않도록 최소 길이로 반영 할 것.
 7. FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 8. 이후 VENDOR 에서 결정한다.
 9. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 결정 예정.
 10. CIF 1st/2nd OXIC TANK는 전회류 포기가 되도록 배관 하여야, 건물외벽에 따른 산기온 설치공간 부족으로 115X PH4 WWT에는 적용 안함.
 11. 1차 호기조 내 AIR 확인이 가능하도록 중간지점에 점검구 반영할 것.
 12. 1차 호기조 내 pH, DO 계측기의 설치 위치는 중간 지점보다 상부에 설치할 것.
 13. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER
 14. LOOSE SUPPLY INSTALLATION BY PURCHASER.
 15. AIR용 조정을 위해 GEAR OPERATED BUTTERFLY VALVE를 적용한다.
 16. 발주가 배관 구성 및 배관 NUMBER는 도면번호 W14-PR-A20-000168을 기준으로 한다.
 17. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치할 것. (TAG NO. 아래 참조)
 18. OPENING SIZE IS 3,500mm x 1,600mm

- 200A-AIR-ST1-N-9518 : PG-W2254
- 200A-AIR-ST1-N-9519 : PG-W2255
- 200A-AIR-ST1-N-9520 : PG-W2256
- 200A-AIR-ST1-N-9521 : PG-W2257

18. OPENING SIZE IS 3,500mm x 1,600mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30

수정	실명	날짜
REV.	DESCRIPTION	DATE
제도	이인학	
검토	한진희	
확인	박동영	
승인	박동영	
제작	사	

SK 에코플랜트

공사명 PROJECT TITLE

M15X Ph4 WWT Project

도면명 DWG. TITLE

PIPING & INSTRUMENT DIAGRAM

CIF MBR ANOXIC & OXIC TANK A

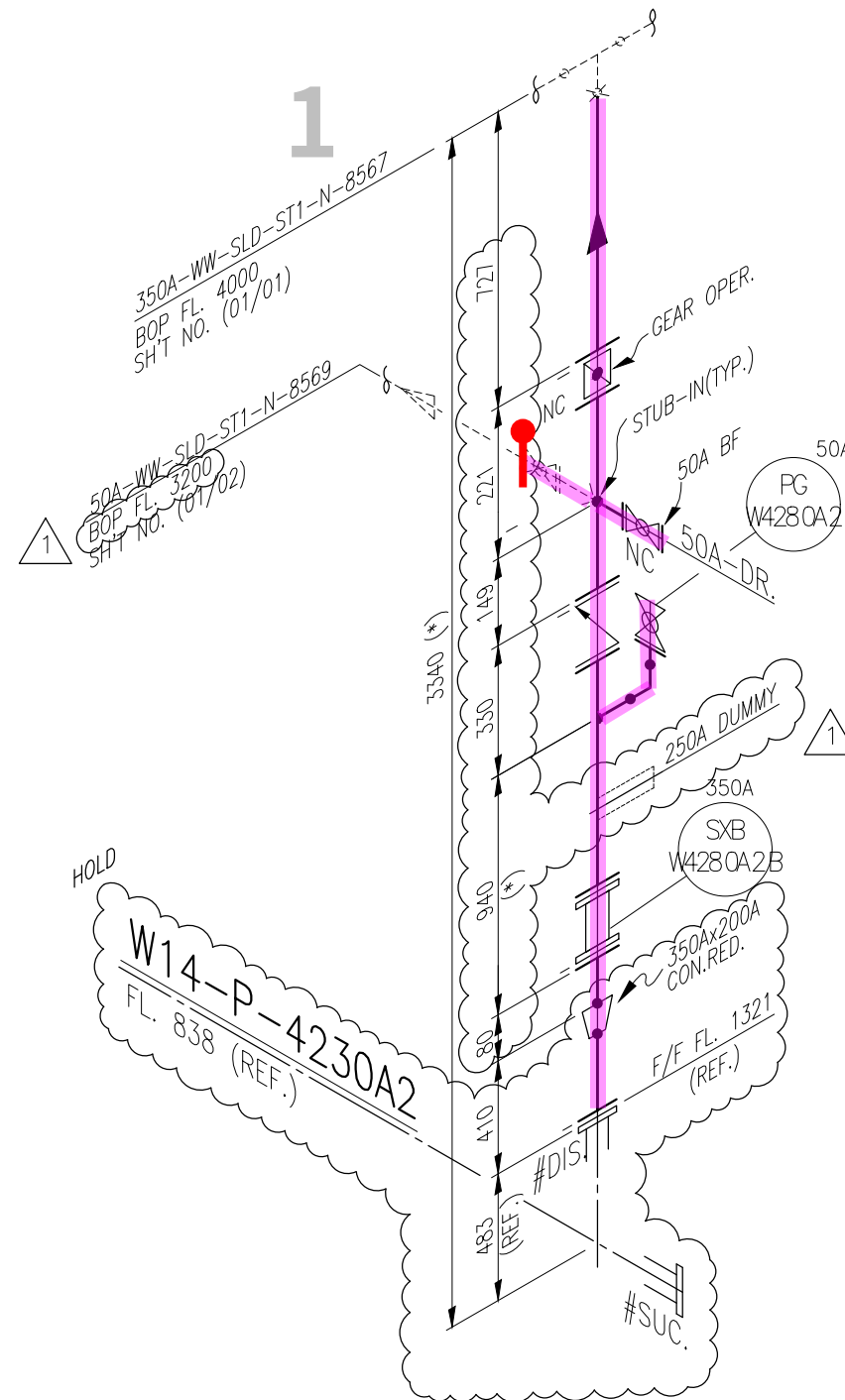
(1/2)

속척 SCALE NONE

날짜 DATE 2025. 08. 14

PROJECT NO. 22330D REV. 1A

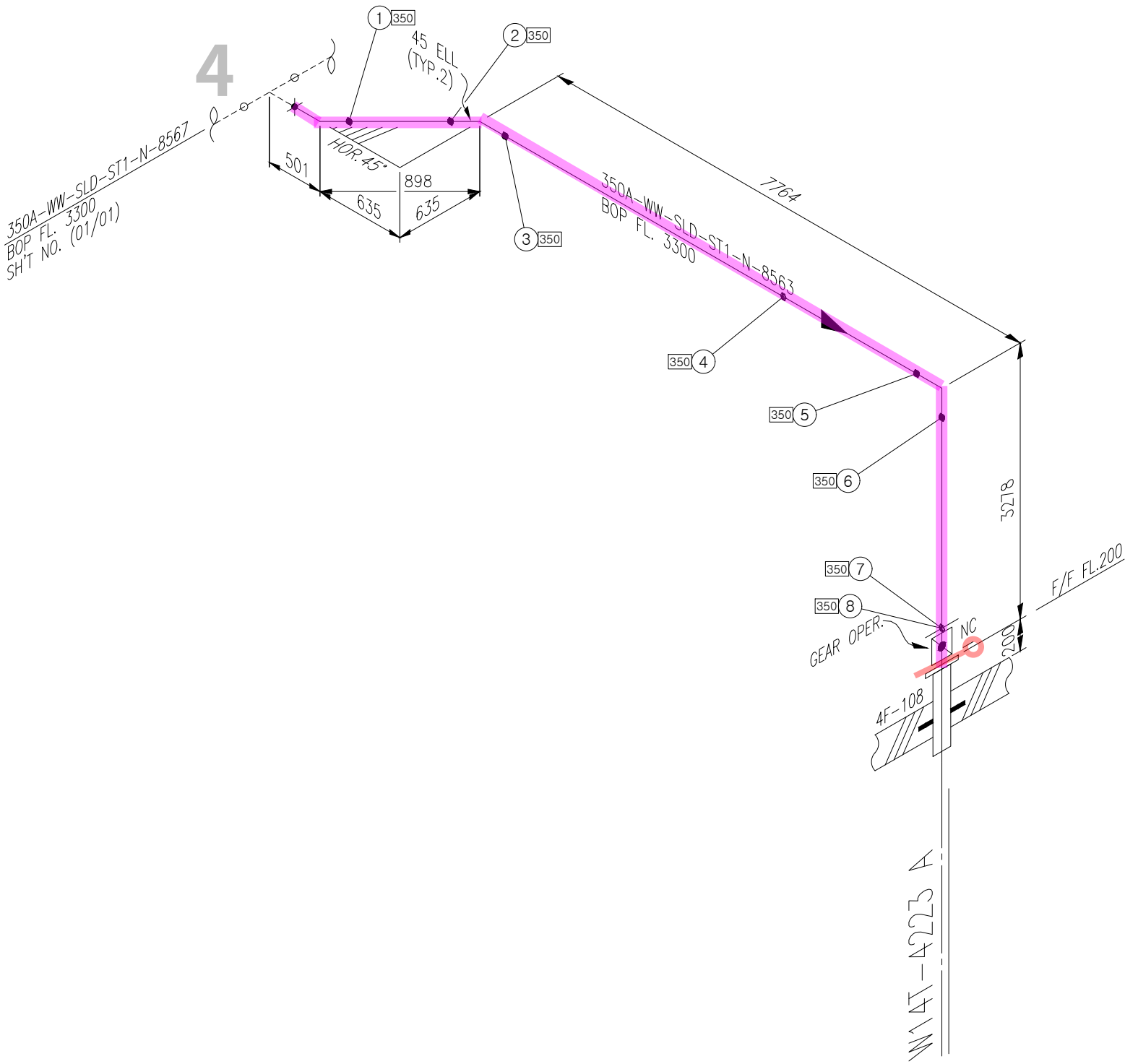
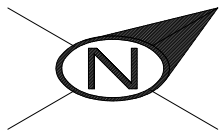
도면번호 SHEET NO. CJ-M15W14-PR-A20-000112 R1



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL-PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

[illegible]

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WW—SLD		22330D	
LINE NO.	SH'T NO.	REV.	
350A-WW-SLD-ST1-N-8568	01 / 01	1	



<NOTE>
1. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	350A	10.4M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	350A	1EA	ELBOW 90 DEG SR, STS304, ERW, BE - SCH 10S
E45	350A	2EA	ELBOW 45 DEG LR, STS304, ERW, BE - SCH 10S
VF	350A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY-GCD450 /
			TRIM-ST304 / EPDM-LINED, GEAR
F	350A	1EA	FLG SO, KS 10K, RF, STS304
G	350A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
BL	350A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)
LOTO	350A	1EA	LOTO

CLIENT

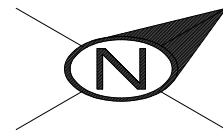


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CONTRACTOR		PROJECT NAME	
SK ecoplant		M15X Ph4 Project	
COMMODITY		JOB NO.	
WW-SLD		22330D	
LINE NO.		SH'T NO.	REV.
350A-WW-SLD-ST1-N-8563		01 / 01	0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14												
	THICK.	-	DESIGN	3.1	60	UNIT	4FA												
			OPERATING	2.0	AMB	REF. PLAN DWG.													
						CJ-M15W14-PI-A30-040002-B													
						REF. P & ID													
						CJ-M15W14-PR-A20-000112													
								△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영						
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D						



CLIENT		 4	
CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY WW—SLD		JOB NO.	
		22330D	
영	LINE NO. 350A—WW—SLD—ST1—N—8567	SH'T NO.	REV.
D		01 / 01	0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : RP * SUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * SUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1														
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	THICK.	-	DESIGN	3.1	60	UNIT	4FA							
			OPERATING	2.0	AMB	REF. PLAN DWG.								
						CJ-M15W14-PI-A30-040002-A								
						REF. P & ID								
					CJ-M15W14-PR-A20-000113	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP			